

WORLD HEALTH ORGANIZATION • CAPSTONE PROJECT

Global HIV Trends & Treatment Effectiveness Analysis

A holistic, data-driven view of the global HIV landscape, 2000–2024

Specialization: Healthcare | Business Focus: Public Health Analysis | Tool: Power BI

Data source: WHO • 150 countries • 6 WHO regions

Project Overview

Over three decades, the WHO and partners have transformed HIV **from a fatal disease into a manageable chronic condition** through prevention programmes, awareness campaigns and the rollout of Antiretroviral Therapy (ART).

Despite this progress, HIV remains a major public-health concern — especially where healthcare infrastructure is limited — and questions persist about the true effectiveness and sustainability of current interventions.

This project evaluates:

- 1 Whether HIV infections are truly declining
- 2 Whether increased treatment access is reducing mortality
- 3 How the overall burden of HIV is evolving globally

Problem Statement & Objectives

The Problem

Despite available HIV data, stakeholders lack a unified analytical framework to:

- Evaluate the effectiveness of treatment programmes
- Understand the link between infections, deaths and treatment
- Identify regions requiring urgent intervention

This limits strategic planning and resource allocation.

The Objectives

Disease monitoring

Track the global spread of HIV over time

Evaluate treatment impact

Assess how ART coverage affects mortality

Support strategic decisions

Help policymakers target high-risk regions

The Data

9,763 annual records • WHO • 150 countries • 2000–2024

New Infections

Newly reported HIV cases — measures disease spread (incidence)

HIV Deaths

Deaths caused by HIV — measures severity & mortality

People Living with HIV

Total individuals living with HIV — measures burden (prevalence)

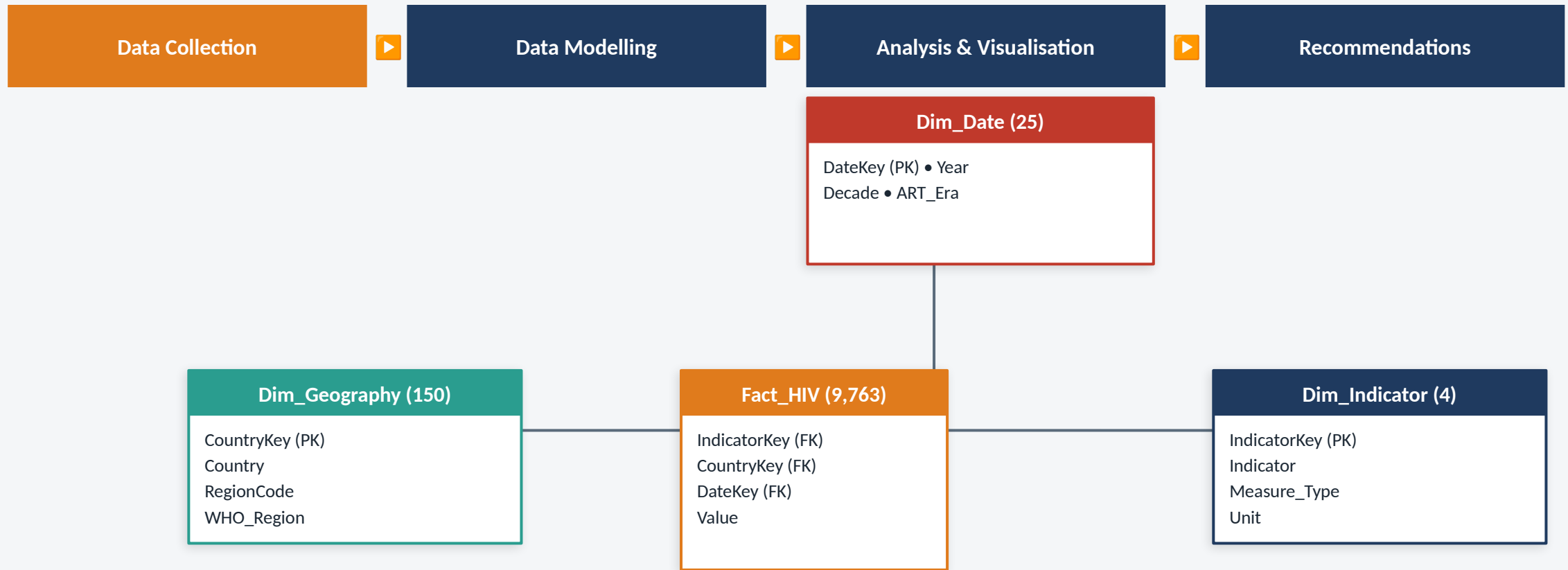
ART Coverage (%)

Share of HIV-positive people on treatment — measures access

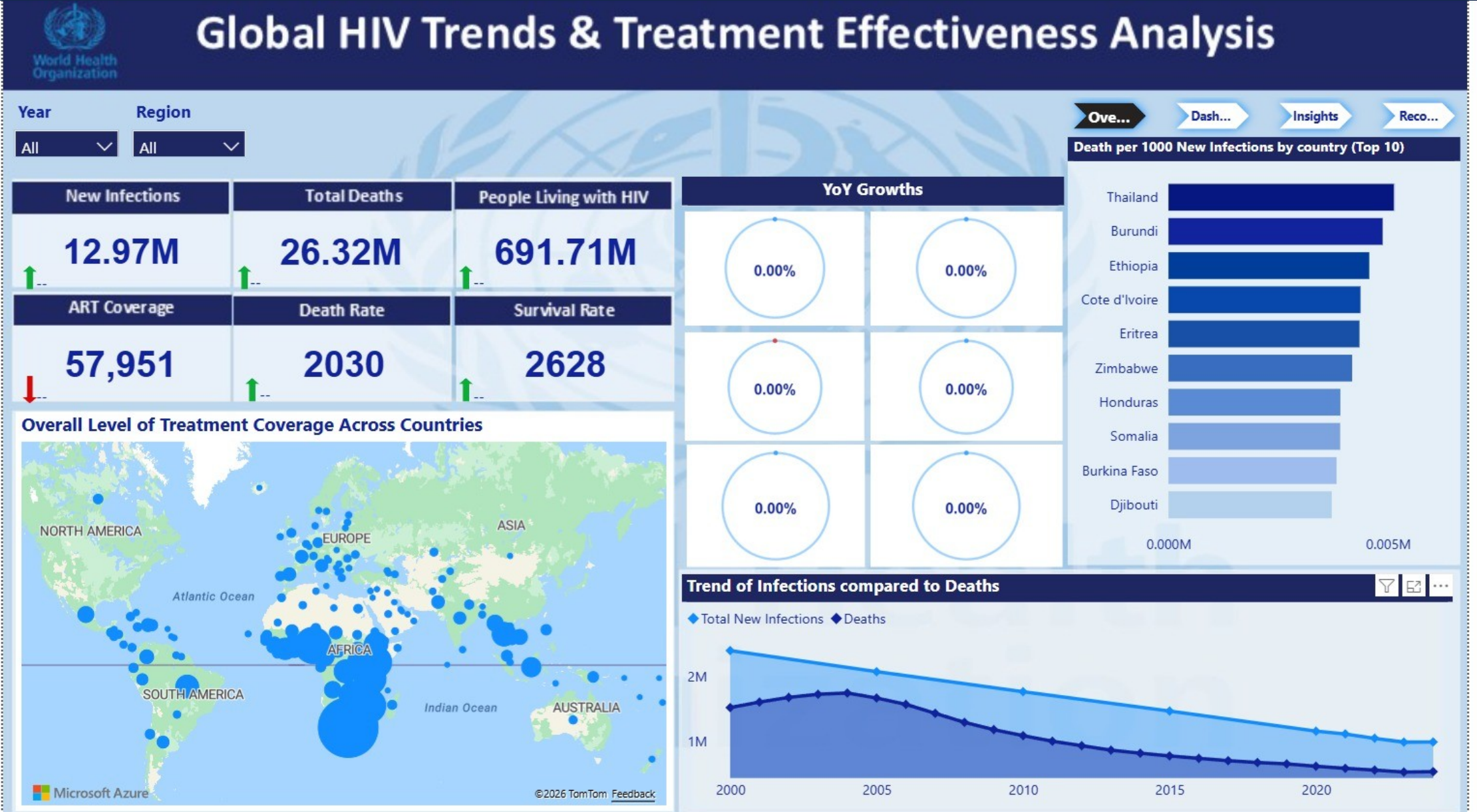
Two data nuances handled in modelling. ART coverage is a percentage (averaged, never summed); and new infections & ART coverage are reported every five years until 2020, then annually — so trends use comparable years rather than treating gaps as zeros.

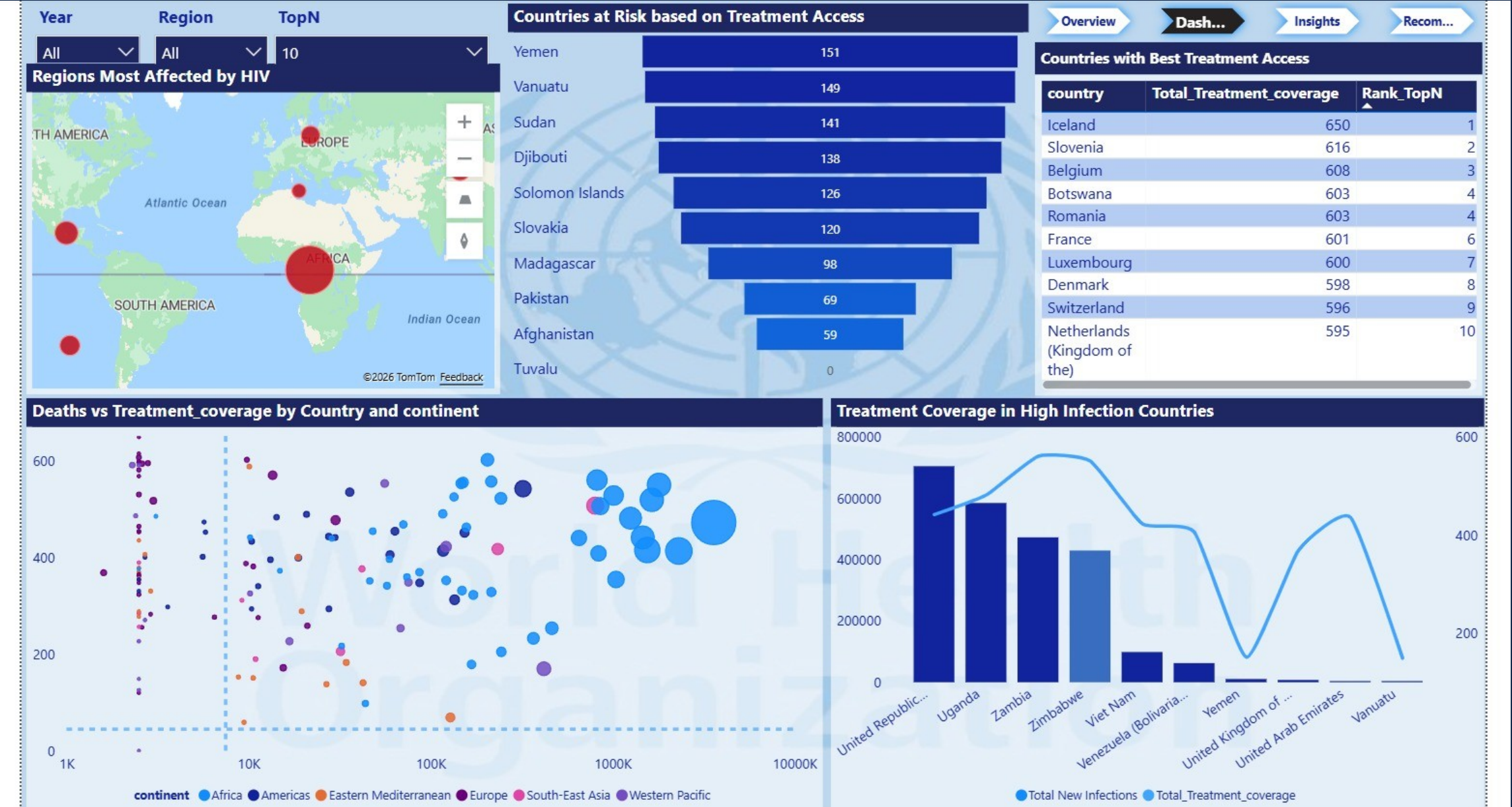
Methodology & Data Model

A clean star schema: 1 fact table + 3 dimensions



KPIs (DAX): Death Rate = Deaths ÷ PLHIV • Survival Rate = 1 – Death Rate • Deaths per 1,000 Infections • ART Coverage (PLHIV-weighted) • YoY Growth





Executive Snapshot

HIV has shifted from a fatal disease to a managed chronic condition — but the job is unfinished.

984K

NEW INFECTIONS

↓ 59% vs 2000

523K

HIV DEATHS

↓ 70% from peak

35.2M

LIVING WITH HIV

rising prevalence

79%

ART COVERAGE

PLHIV-weighted

1.49%

DEATH RATE

deaths ÷ PLHIV

1

Infections down, but stalling

Down 59% since 2000 — yet essentially flat from 2023 to 2024 (+0.25%).

2

Treatment is working

ART coverage and death rate move in opposite directions ($r \approx -0.48$).

3

Burden keeps growing

More people live with HIV each year, so the care burden expands.

Figures reflect the most recent reporting year (2024); the dashboard's cumulative card sums prevalence across all years.

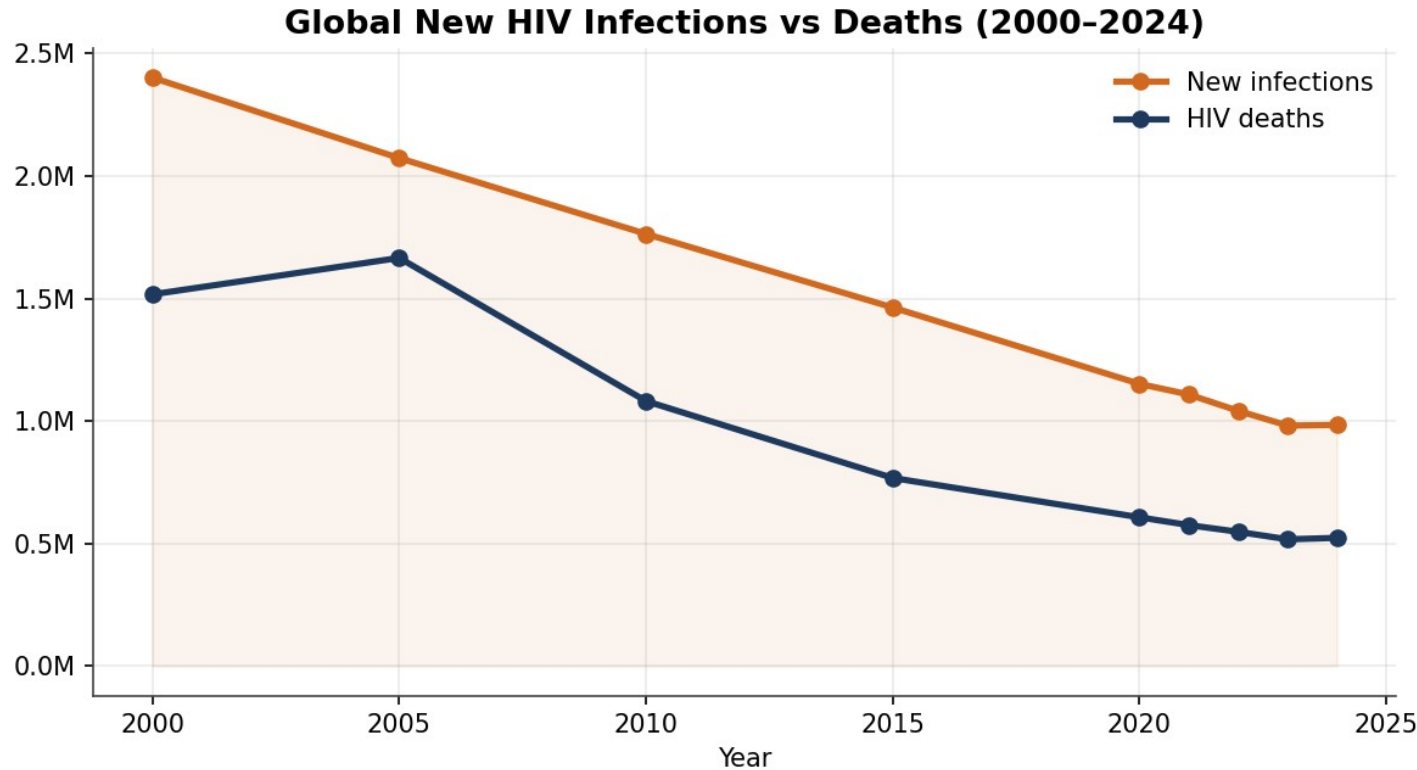
Business Questions Answered

Every question in the project brief, addressed by the dashboard.

- 1 How many new infections over time?**
Fell 59% — 2.40M (2000) → 0.98M (2024); now plateauing.
- 2 Total HIV-related deaths globally?**
~523K in 2024; down ~70% from the 2004 peak.
- 3 How many people living with HIV?**
~35.2M in the latest year and rising (chronic-condition effect).
- 4 Overall treatment coverage?**
~66% average / ~79% PLHIV-weighted ART coverage (2024).
- 5 Mortality & survival rate?**
Death rate $\approx 1.49\%$; survival rate $\approx 98.51\%$.
- 6 Highest deaths per 1,000 infections?**
Lesotho, Ethiopia, Malawi, Zimbabwe, Thailand.
- 7 Infections vs deaths trend?**
Both falling; deaths have fallen faster than infections.
- 8 Most affected regions?**
Africa (~75% of burden); top death rates in E. Mediterranean & W. Pacific.
- 9 Best treatment access?**
Rwanda, Botswana, Namibia, Zimbabwe, Eswatini, Zambia (90%+).
- 10 Do high-infection countries have cover?**
Mostly yes (S. Africa, Nigeria 80%+) — but Indonesia 41%, Philippines 40% lag.
- 11 ART coverage vs mortality?**
Negative correlation ($r \approx -0.48$): more coverage, lower mortality.
- 12 Most at-risk countries?**
Pakistan 16%, Sudan 20%, Madagascar 29%, Congo 34%, Indonesia 41%.

Are infections truly declining?

Both new infections and deaths have fallen sharply — but the decline is flattening.



What the data says

2.40M → 0.98M new infections

a 59% drop since 2000.

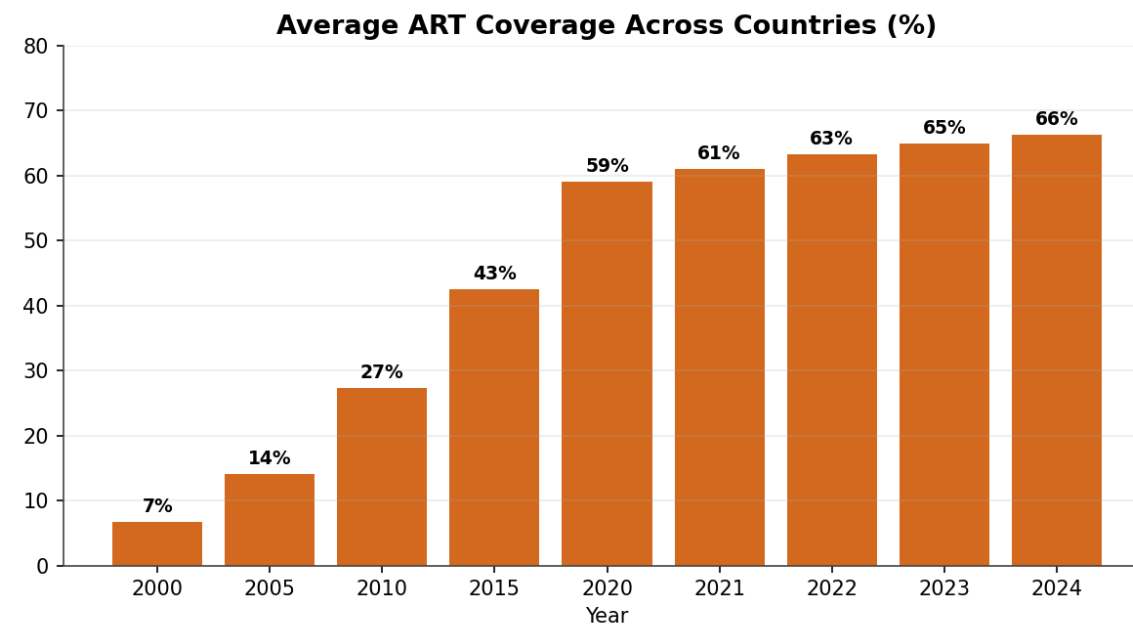
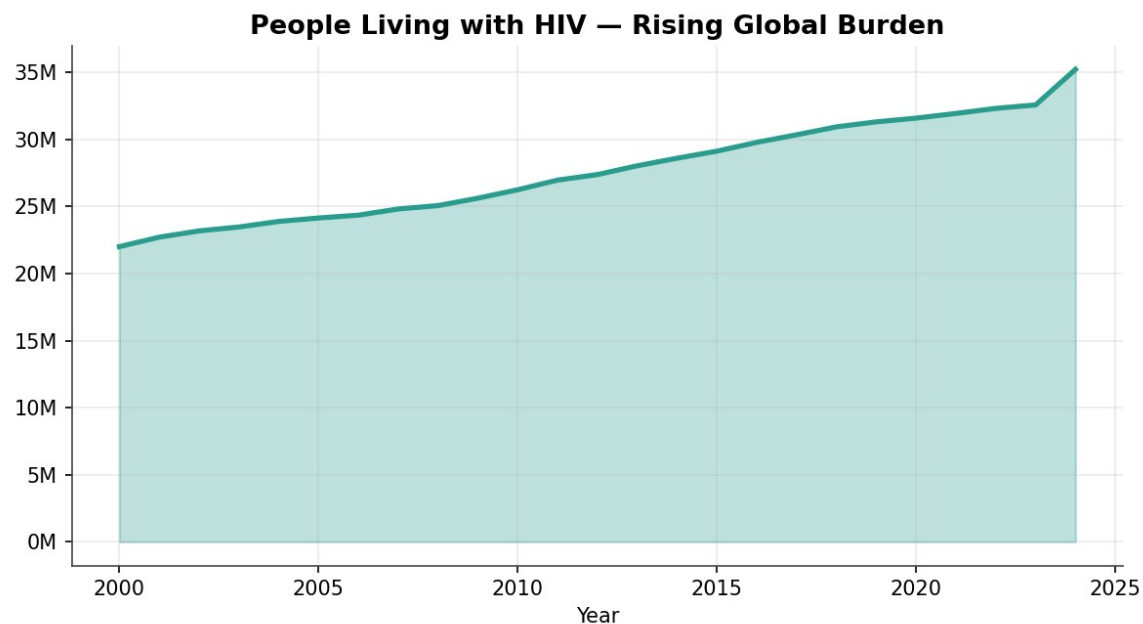
Deaths down ~70% from the 2004 peak.

BUT 2023→2024 was almost flat (+0.25%).

Prevention gains are plateauing — renewed effort is needed to keep the curve falling.

A success that creates a new challenge

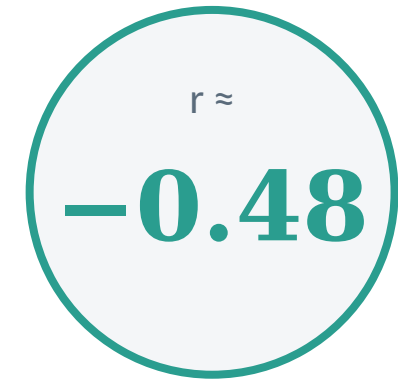
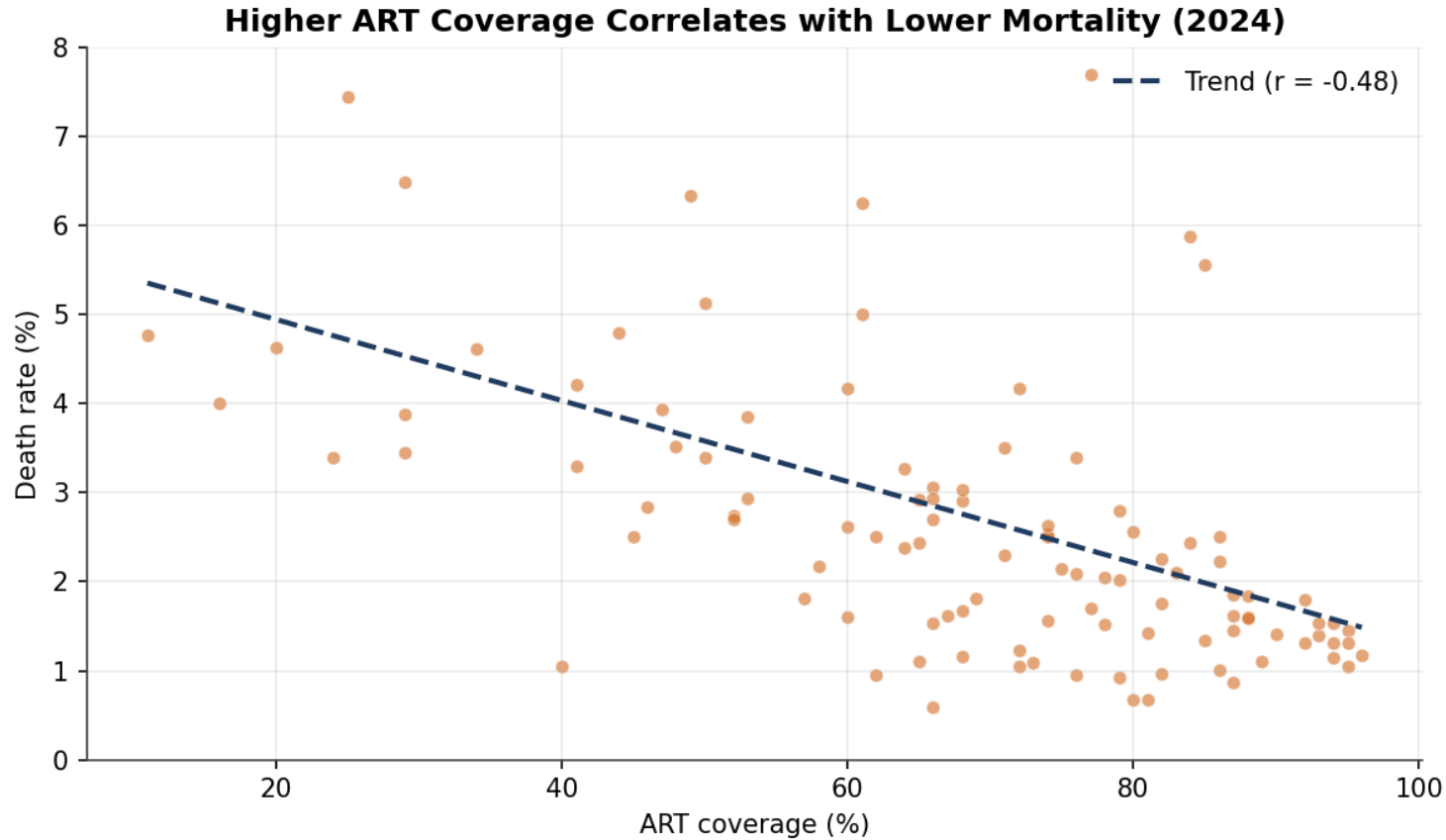
As deaths fall, more people live with HIV — driven by a ten-fold rise in treatment access.



The chronic-condition signature. People living with HIV climbed from 22M (2000) to 35.2M (2024) because ART now keeps people alive — average coverage rose from 7% to 66%. The mortality win is real, but it locks in a permanent, growing need for lifelong treatment and financing.

Does treatment reduce mortality?

Across 108 countries, higher ART coverage is associated with lower death rates.

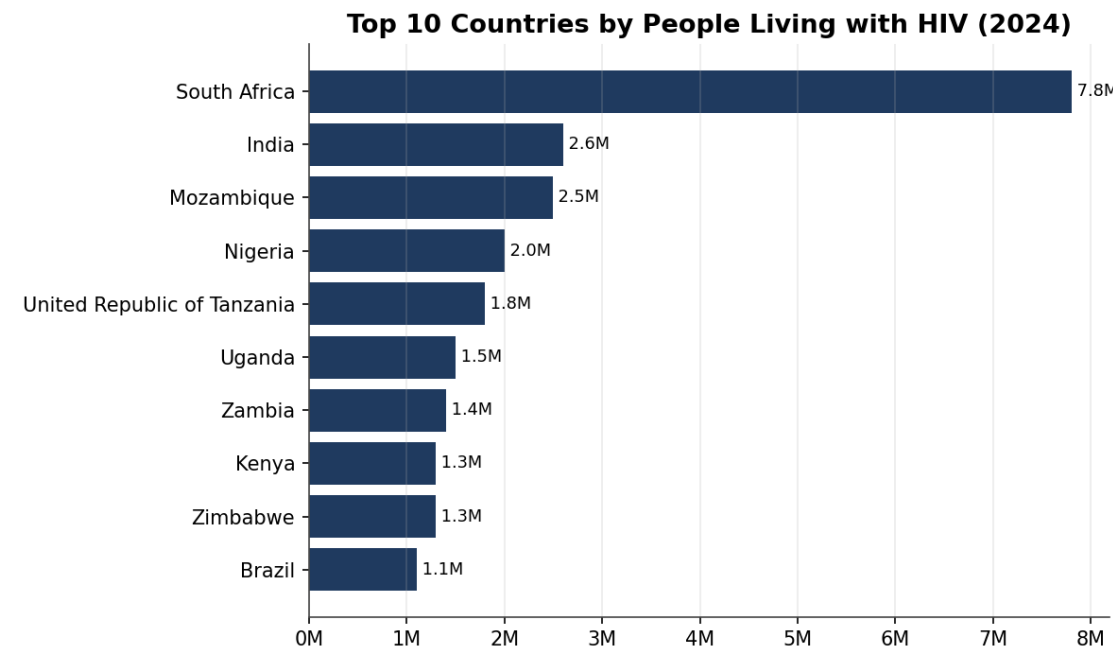
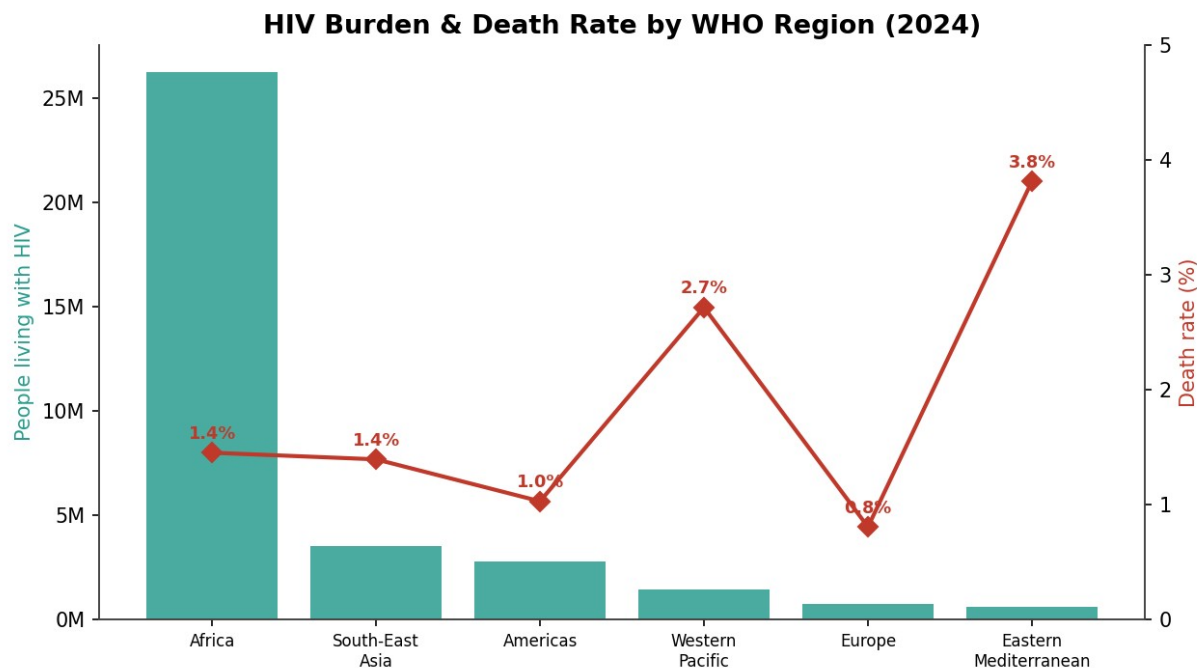


negative correlation

This is the core evidence that treatment programmes save lives. The link is moderate — outcomes also depend on early diagnosis, adherence and health-system quality — but the direction is unmistakable.

The geography of the epidemic

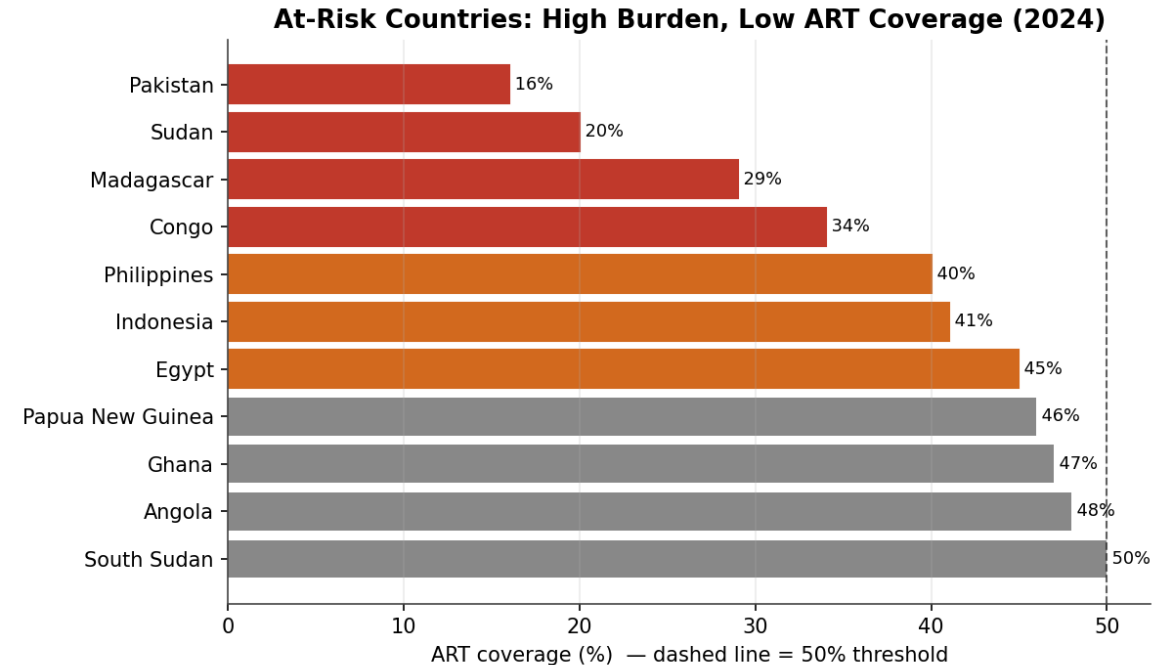
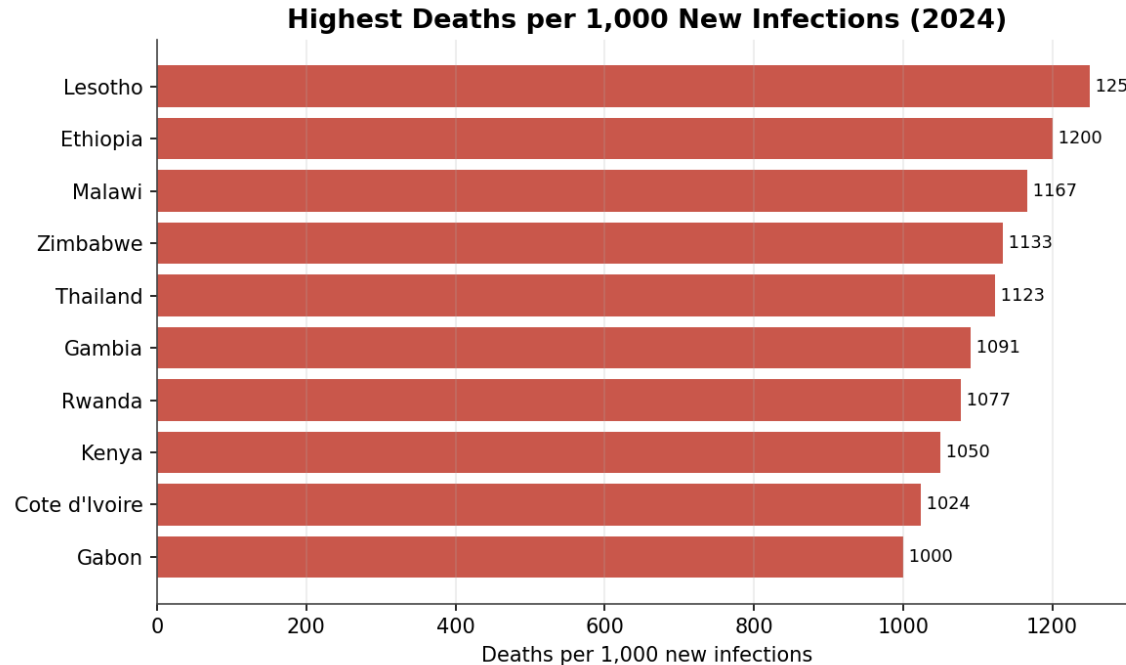
Africa carries ~75% of the burden — but the highest death rates sit elsewhere.



Burden ≠ death rate. Africa leads on absolute burden (26.3M living with HIV, led by South Africa at 7.8M). Yet the Eastern Mediterranean (3.81%) and Western Pacific (2.71%) post the highest death rates — regions where treatment access lags the size of the epidemic.

Where intervention is most urgent

Two lenses: where deaths outnumber new infections, and where coverage is dangerously low.



The priority list. Lesotho, Ethiopia, Malawi and Zimbabwe record more deaths than new infections (>1,000 per 1,000) — a retention-in-care problem. Pakistan (16%), Sudan (20%), Madagascar (29%) and Indonesia (41%) pair large burdens with sub-50% ART coverage — the clearest targets for urgent scale-up.

Dashboard Insights

What the analysis tells us about burden, treatment and implementation.

Dashboard Insight

- ▶ Across 2000–2024 the dashboard counts 690M+ people living with HIV. Rising prevalence isn't worsening infections — better ART keeps patients alive, so HIV is increasingly a manageable chronic condition. Higher prevalence can reflect healthcare success.
- ▶ African countries carry the highest HIV burden — high infection rates, large numbers living with HIV and significant mortality — reflecting how healthcare inequality, poverty and limited treatment access affect outcomes across the region.
- ▶ In many countries death rates are falling faster than infection rates — evidence that healthcare systems and HIV interventions are improving survival through better access to medication and long-term care.

Dashboard Insight

- ▶ Countries with stronger ART coverage generally show lower mortality, better survival and improved long-term disease management. As treatment access improves, mortality tends to fall.
- ▶ Yet some countries record high “Deaths per 1,000 New Infections” even where coverage looks moderate or high — pointing to weak healthcare delivery, late diagnosis, poor follow-up, medication interruptions or data-reporting gaps.
- ▶ Treatment availability alone is not enough if implementation systems are weak — strong delivery, monitoring and reporting are just as critical as access.

Recommendations

✓ **Expand access, prevention & early diagnosis**

Reducing the global HIV burden requires early diagnosis, improved treatment access, stronger healthcare systems and effective prevention. Governments and health organizations should expand community testing, awareness campaigns and mobile screening to encourage early detection and reduce transmission. Increasing access to affordable ART — particularly in high-burden regions such as Sub-Saharan Africa — is essential to lowering HIV-related deaths and improving quality of life.

✓ **Strengthen systems, monitoring & data**

Healthcare systems must strengthen patient monitoring, treatment distribution and data-reporting to ensure consistent care and informed decisions. Preventive education, condom distribution and youth-awareness initiatives remain critical to reducing new infections. Tools such as Microsoft Power BI help monitor trends, identify high-risk populations and support evidence-based policymaking. Sustainable progress depends on integrating prevention, treatment, accurate reporting and data-driven strategies.

Priority Actions

Five data-driven moves, ranked by impact.

1 Close the coverage gaps

Fund low-ART, high-burden countries — Pakistan, Indonesia, Madagascar, Congo, Sudan.

2 Re-energise prevention

Renew PrEP, testing & education as new infections plateau — focus on Africa & the Philippines.

3 Fund the growing care pool

Sustainable, long-term financing for lifelong ART so mortality gains aren't reversed.

4 Target retention hotspots

Lesotho, Ethiopia, Malawi, Zimbabwe — early diagnosis, adherence & viral suppression.

5 Modernise data reporting

Move five-yearly indicators to annual reporting for timelier course-correction.

The bottom line

A public-health success in motion — with unfinished business.

Infections down 59%, deaths down 70% from peak, treatment up ten-fold. Sustained, targeted, data-driven investment — guided by the dashboard and model built in this project — is what keeps the curve falling.

Built with Power BI • Data: World Health Organization • 2000–2024